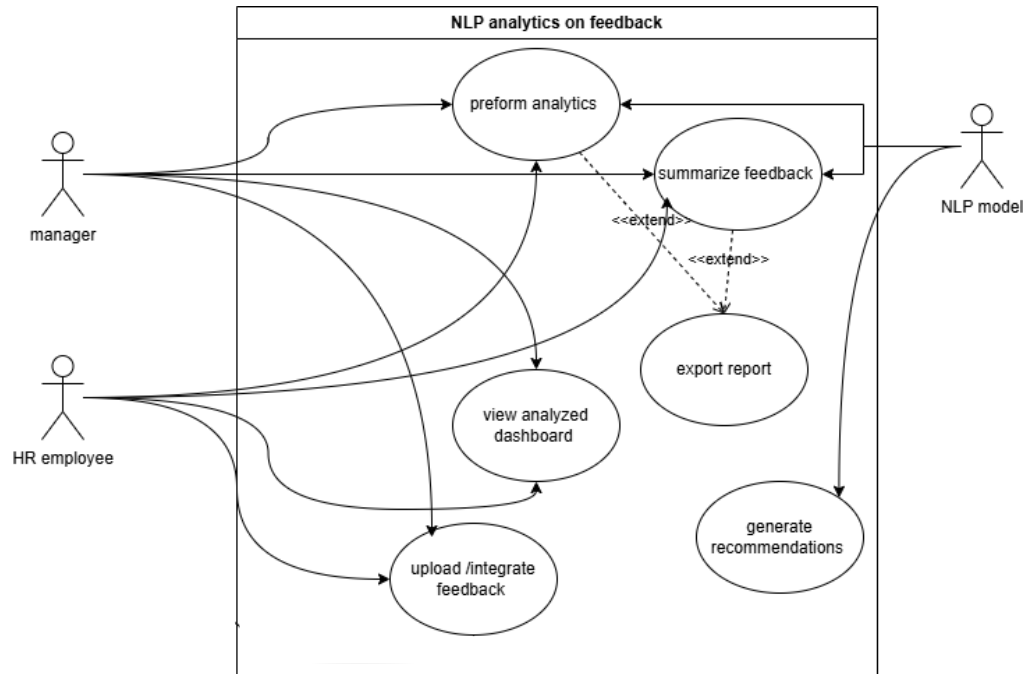


3 Framework

3.1 Business Process

This feature aims to perform NLP analysis techniques on the feedback, so it generates the summarization and the analytics on the feedback. The system gets the feedback from either the direct upload or by integration with the evaluation system, then the user chooses what to be done: 1- generate a summary 2- perform analytics on the feedback, or both. then, the internal dashboard gets updated with the analyzed data. Finally user choose to export the result or not.

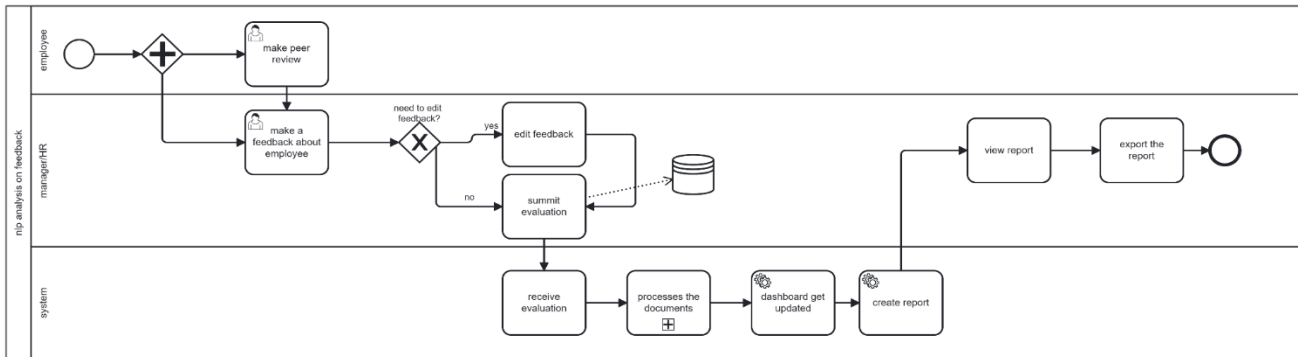
3.1.1 Use Cases



(Figure 3, use case diagram)

The use-case diagram shows how managers, HR staff, and employees interact with an NLP-based system to upload feedback, analyze it, view results, and generate reports. The NLP model supports automated tasks such as summarizing feedback, performing analytics, and producing recommendations. Overall, the diagram outlines the full workflow of transforming raw feedback into actionable insights through NLP.

3.1.2 Business Process Diagram (BPMN)



(figure 4, Business Process Diagram)

The BPMN diagram shows the workflow of : employees submit reviews, managers/HR edit and confirm them, and the system processes the input to update dashboards and generate reports. The decision step for editing feedback is correctly placed, and the process ends with viewing and exporting the final report

3.1.3 User Stories

ID	category	As a...	I want to...	So that...
1	feedback generation, integration	HR employee	write employee feedback text into the system	all peer and manager evaluations are stored in one place for analysis
2	feedback generation, integration	Manager / HR	submit a completed evaluation/ feedback	the system can include it in the overall performance analysis
3	feedback generation, integration	Manager / HR	review and edit my feedback before submitting	I can correct mistakes and ensure my comments are clear and fair
4	system analysis on the feedback	HR / Manager	trigger NLP analytics on the collected feedback	the system can automatically detect sentiment, key topics, and patterns
5	system analysis on the feedback	HR / Manager	see a dashboard of analyzed feedback results	I can quickly understand trends, strengths, and issues across employees or teams
6	system analysis on the feedback	Manager / HR	view a summarized report for a specific employee	I can discuss performance using clear, concise insights instead of raw comments
7	result export	Manager / HR	receive system-generated recommendations based on the analyzed feedback	I can plan development actions, coaching, or corrective measures for employees
8	Meta data	Manager / HR	Create an employee with credentials and a department	I can store the employees and follow up on the same employee
0	Meta data	Manager / HR	Select an employee from the dropdown menu	I make a review thread on the same employee

(Table 2, user stories)

Field Requirements / Business Rules:

Field ID	Field type	Field Explanation
FA-FR-02	Field Requirement	Paste text box supports multi-line and minimum character validation.
FA-FR-03	Field Requirement	Analyze button enabled only when input is valid.
FA-FR-04	Field Requirement	Recent analyses show name, timestamp, score, sentiment and result.
FA-BR-01	Business Rule	Either file or text required for analysis, not both empty.
FA-BR-03	Business Rule	Only HR/Managers permitted to run analysis.
FA-BR-04	Business Rule	Recent analyses sorted by newest first.
FA-BR-05	Business Rule	Sentiment labels follow predefined categories.
DA-FR-01	Field Requirement	Display AI summary, sentiment, score, confidence.
DA-FR-02	Field Requirement	Sentiment distribution chart with percentages.
DA-FR-03	Field Requirement	Emotion detection chart shows intensities.
DA-FR-04	Field Requirement	Topic modeling shows key themes with frequency counts.
DA-FR-05	Field Requirement	Evaluation alignment radar chart compares reviewer groups.
DA-FR-06	Field Requirement	Strengths, development areas, and recurring themes displayed.
DA-BR-02	Business Rule	Negative keywords highlighted visually.
DA-BR-03	Business Rule	Sentiment distribution must sum to 100%.
DA-BR-04	Business Rule	Missing evaluator scores default to 'N/A'.
DA-BR-05	Business Rule	Overall score calculated from weighted sentiment/themes.

(Table 3, Field Requirements)

Wireframe:

<https://yoyo-chant-25512223.figma.site>

Test Cases:

ID	Test Scenario	Test Steps	Expected Result
1	write valid feedback	1) Log in as HR employee. 2) Navigate to Feedback write feedback. 3) Select a employee 4) Click Upload.	File is uploaded successfully, records are stored, a confirmation message appears.

2	Employee submits a peer review	1) Log in as Employee. 2) Navigate to Peer Reviews. 3) Choose a colleague from the assigned list. 4) Enter textual feedback. 5) Click Submit.	Peer review is saved with status “Submitted”, visible to HR/Manager in the evaluation list, and timestamped with the employee ID.
3	Trigger NLP analytics on a feedback batch	1) Log in as HR / Manager. 2) Ensure at least one batch of feedback is available. 3) Click Run Analysis.	Job is queued/processed, topics, and summaries are generated and stored.
4	View analyzed feedback dashboard	1) Log in as HR / Manager. 2) Navigate to Analytics → Dashboard. 3) Team/department filter. 4) Apply filter.	Dashboard displays updated charts and KPIs (e.g. sentiment distribution, top topics, volume of feedback) for the selected filters with no errors. And recommendations for the development
5	Export analysis report to Excel/PDF	1) Log in as HR / Manager. 2) Open any completed analysis report. 3) Click Export → Excel/PDF. 4) Choose save location in the browser dialog.	An .xlsx / .pdf file is downloaded, containing data.
6	Login validation	1)Click on register 2)fill the fields 3)click submit	Credentials are registered and directed to login page
7	Dashboard visuals	1) Navigate to the dashboard 2) Click on filter	Filters are applied to the statistical representation of the data
8	Create employee	1)navigate to reviews 2)click on create employee 3)fill employee data 4)click create	The employee will be created and his name is displayed at the drop-down

(Table 4, test case)

4 Methodology

4.1 Overview of research methodology

A robust research methodology serves as the foundational blueprint for any scientific investigation, saying the actions used to collect, analyze, and interpret data. So, an appropriate framework is needed for validity, reliability, and practical relevance of the findings. The following provides a comprehensive overview of four methodological approaches: Data Science, Design Science, Behavioral, and Case Study.

Data science methodologies provide essential structured frameworks for extracting valuable, hidden insights from massive and complex datasets. Knowledge Discovery in Databases (KDD) is defined as the non-trivial process of identifying valid, novel (Nwagu et al., 2017). The KDD process requires human interaction and domain